

HART-120

Flexible High-Aspect-Ratio Wing Design & Robust Gust-Load Alleviation

Reduced-Order Aero-Structural, Control, Mission and ML-Assisted Design Study

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Original fictional aircraft concept. No proprietary OEM geometry, loads or control laws are used.

Executive Summary

Industrial question. Can a higher-aspect-ratio transport wing deliver a meaningful aerodynamic and mission benefit after the structural mass, aeroelastic flexibility, maneuver/gust loads, load-alleviation hardware and robustness requirements are all accounted for?

Approach. HART-120 was developed as an original fictional 120-passenger-class concept so the engineering workflow could be demonstrated without relying on proprietary aircraft data. The model couples spanwise aerodynamics, beam/wingbox sizing, static aeroelastic load redistribution, active maneuver and gust-load alleviation, fail-safe web sizing, mission-level fuel accounting, machine-learning surrogates and Monte Carlo uncertainty analysis.

Final project result. Within the reduced-order model and project-defined constraints, the final AR 15.5 technology configuration closes all final design gates. The aircraft-level model predicts 8.36% lower induced drag, 2.28% lower total cruise drag, a 4.97% increase in L/D and a 2.01% cruise-fuel saving on the 3000-km study mission. The final preview-based gust-load-alleviation controller reaches 98% joint robustness over the final 300-case uncertainty assessment.

Metric	Baseline / Requirement	Final result
Aspect ratio	13.5	15.5
Wing span	39.13 m	41.93 m
Aircraft mass	55,000 kg	56,416.5 kg
Final structural + control package	-	6,218.69 kg
Structural budget	6,242.85 kg	24.16 kg margin
Induced-drag reduction	0%	8.36%
Total cruise-drag reduction	0%	2.28%
L/D	18.91	19.85 (+4.97%)
3000-km cruise fuel	6,234.17 kg	6,109.06 kg
Joint GLA robustness	>=95%	98.0%
P95 controlled root moment	<=3.729 MN·m	3.687 MN·m
Maximum actuator deflection	<=15 deg	13.89 deg

Engineering conclusion: the project demonstrates a viable conceptual path to a moderately higher-AR wing, but only when aeroelastic tailoring, active load alleviation, fail-safe structure, control-system mass and uncertainty are treated as one coupled design problem.

1. Project Scope and Claim Boundary

The HART-120 study intentionally addresses the engineering decision process rather than claiming a certified aircraft design. The project began with the aerodynamic incentive for higher aspect ratio and was progressively extended whenever a new physical or systems-level constraint threatened the apparent benefit.

- Baseline planform and elliptical reference loading.
- Flexible-wing stiffness calibration and simplified spar-cap/web sizing.
- Material-redistribution sensitivity and mass-efficiency study.
- 2.5-g maneuver and vertical-gust load cases.
- Active maneuver/gust load alleviation with control-authority limits.
- Actuator lag, delay, rate limits and surface-deflection constraints.
- First-mode dynamic gust response and gust-gradient sensitivity.
- Static aeroelastic twist and passive load redistribution.
- Compression-panel buckling and support-spacing screening.
- Aspect-ratio trade from AR 13.5 to 18.
- Fail-safe structural sizing under load-alleviation degradation.
- Aircraft-level drag, L/D and cruise-fuel closure.
- Physics-generated ML surrogates for fast design screening.
- Full-physics re-verification of ML-selected candidates.
- Monte Carlo robustness assessment and final controller redesign.

1.1 What this report can claim

The results support comparative, project-level conclusions within the adopted model assumptions: how AR changes induced drag and structural burden; where stiffness is most effective; how passive and active load alleviation alter selected root-load metrics; how fail-safe requirements and system mass erode the benefit; and whether a selected control architecture remains acceptable under the chosen uncertainty ranges.

1.2 What this report does not claim

The study is not a substitute for high-fidelity aerodynamic analysis, nonlinear aeroelasticity, detailed composite wingbox FEA, flutter clearance, CS/FAR-25 certification substantiation, flight-control-law certification, actuator thermal/reliability qualification or complete aircraft mission analysis.

2. Baseline Wing, Loads and Flexible Structure

The fictional baseline aircraft uses a design mass of 55,000 kg, Mach 0.76 cruise at 11 km, $CL = 0.52$, $AR = 13.5$ and taper ratio 0.30. These requirements produce a 113.42 m^2 wing and 39.13 m span. The baseline span is discretized into 51 half-wing stations.

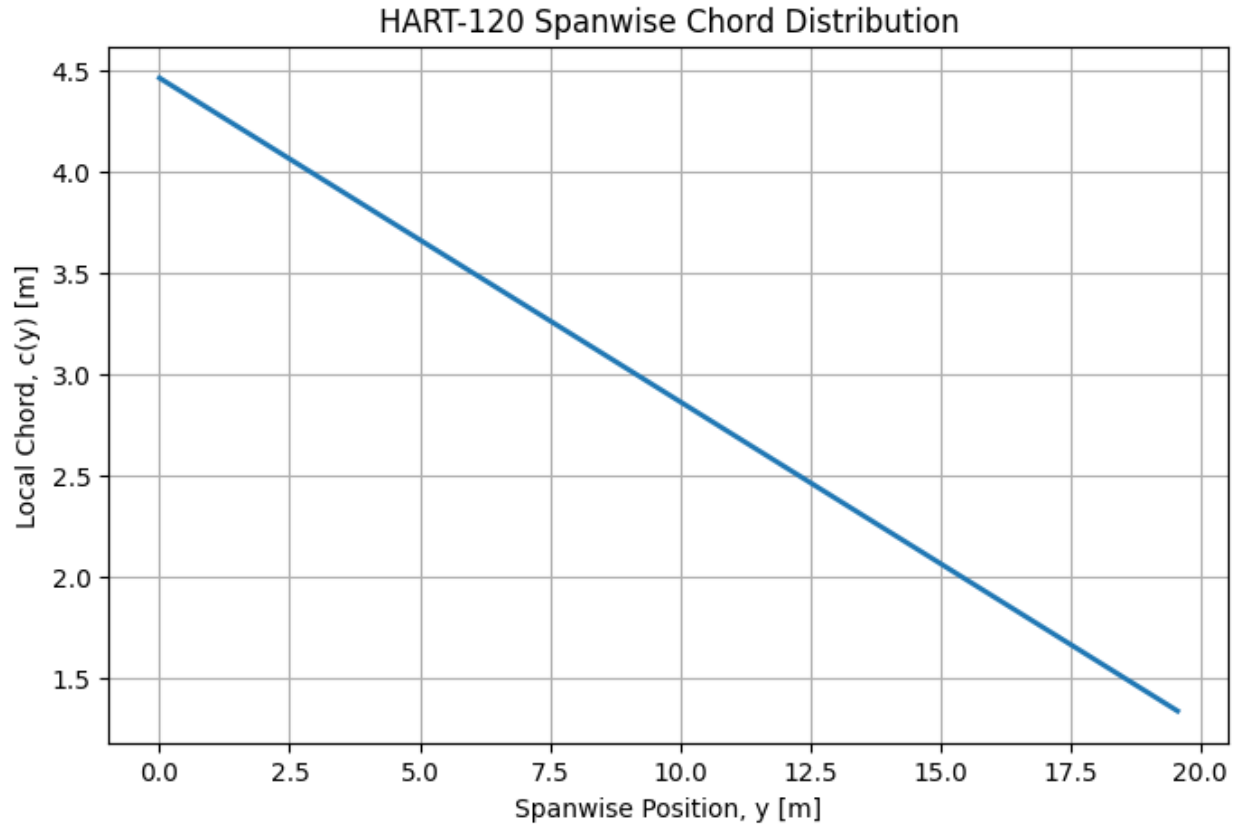


Figure 1. Baseline HART-120 trapezoidal chord distribution.

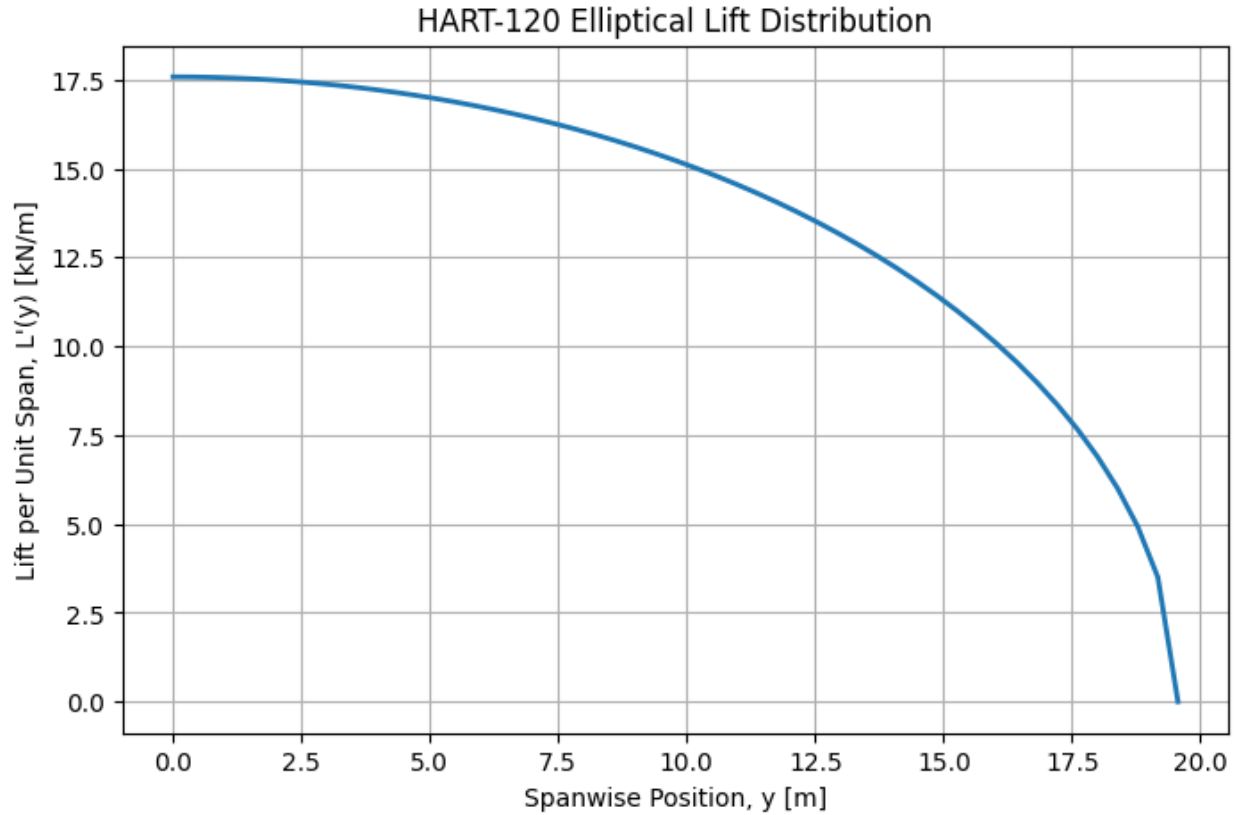


Figure 2. Elliptical reference lift distribution used for the initial 1-g cruise load model.

The reference lift distribution is integrated spanwise to obtain shear force and bending moment. The baseline half-wing carries 269.77 kN, and the 1-g root bending moment is 2.237 MN·m.

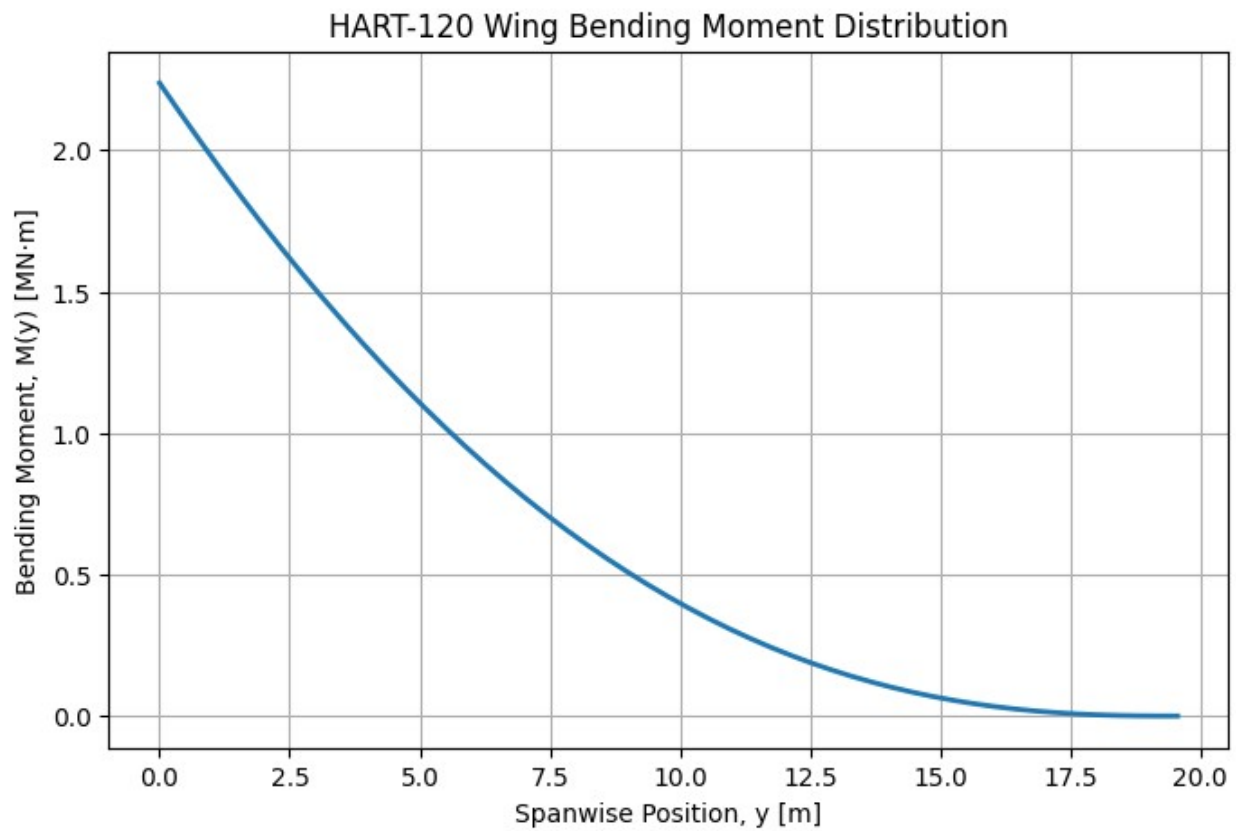


Figure 3. Baseline 1-g bending-moment distribution.

2.1 Flexible-wing calibration

A reduced-order Euler-Bernoulli model is used to represent global bending. The prescribed stiffness shape follows chord³ and the root EI is calibrated so the baseline 1-g tip deflection matches a project-level target equal to 5% of half-span. This is a calibration target, not a certification limit.

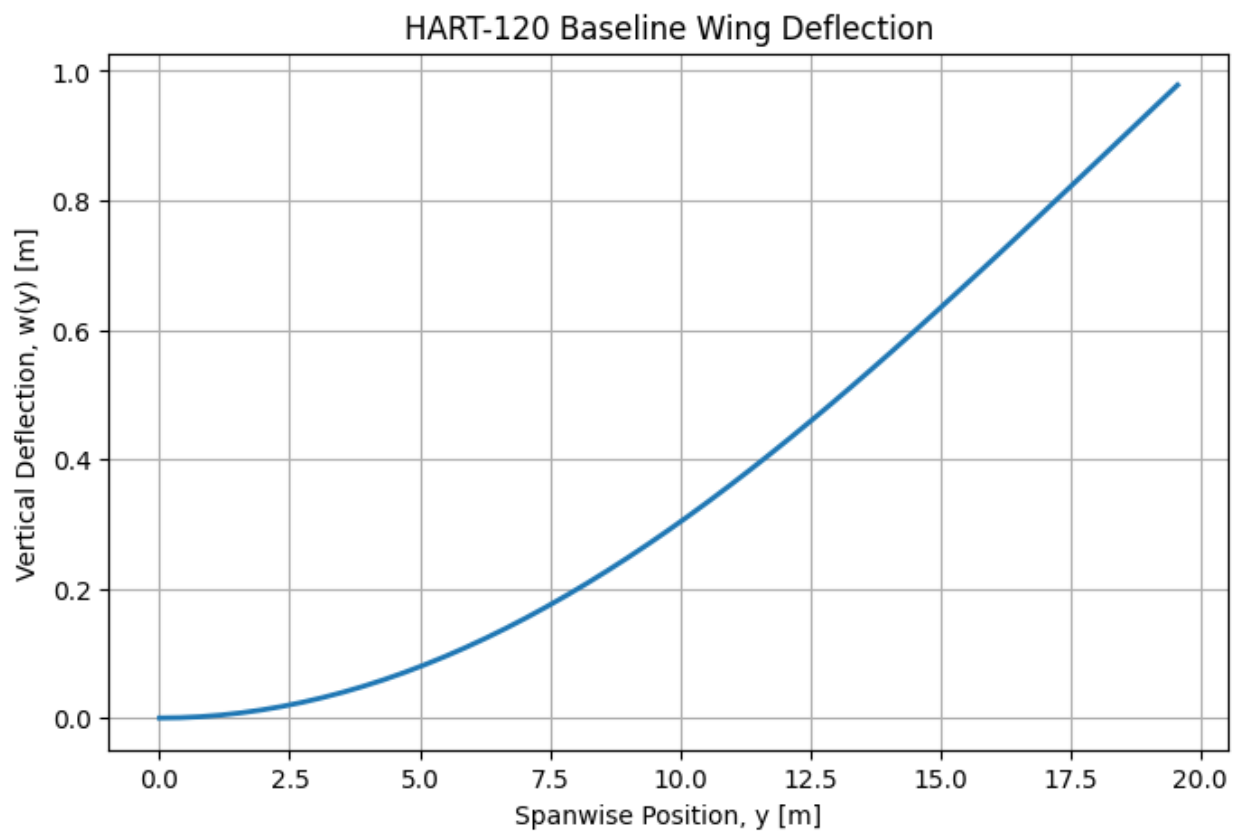


Figure 4. Calibrated baseline flexible-wing deflection.

3. Simplified Wingbox and Structural Efficiency

The conceptual wingbox is represented by two equivalent spar caps separated by the box height plus two shear webs. An equivalent modulus of 70 GPa and density of 1600 kg/m³ are used. The model is deliberately simplified; reported wingbox mass refers to spar caps + webs only.

The initial stiffness-driven design produces a simplified wingbox mass of approximately 4.54 t. Regional stiffness-placement studies then test whether the same structural mass can be used more effectively before increasing total mass.

Equal-mass placement finding: adding the same 100 kg per half-wing to the inner region reduced tip deflection about 10.7 times more effectively than placing it in the outer region.

A fixed-mass redistribution of 100 kg per half-wing from the outer region to the inner region reduced tip deflection from 0.9782 m to 0.9275 m while preserving cap mass. The outboard local stress increased, demonstrating the expected trade between global stiffness efficiency and local structural utilization.

3.1 Maneuver and gust loading

The design is then subjected to a 2.5-g maneuver and vertical gust loading. These load cases expose a major limitation of the initial 1-g stiffness sizing: strength and shear requirements can become governing when the wing is sized for realistic off-design loads.

4. Load Alleviation, Actuator Dynamics and Aeroelasticity

Active load alleviation is modeled as zero-net-lift spanwise redistribution: load is shifted from outboard to inboard regions so the aircraft lift requirement is preserved while root bending moment is reduced. Control demand is then converted into an equivalent local change in lift coefficient and surface-deflection requirement.

The control model is extended to include actuator first-order response, processing/sensor delay, rate limiting and a 15-degree project surface-deflection limit. A first bending mode is used for reduced-order dynamic gust response.

4.1 Passive aeroelastic load redistribution

Static torsional deformation creates washout under load. The resulting passive aeroelastic redistribution reduces outboard lift and transfers load inboard, lowering the root bending demand before active control is applied.

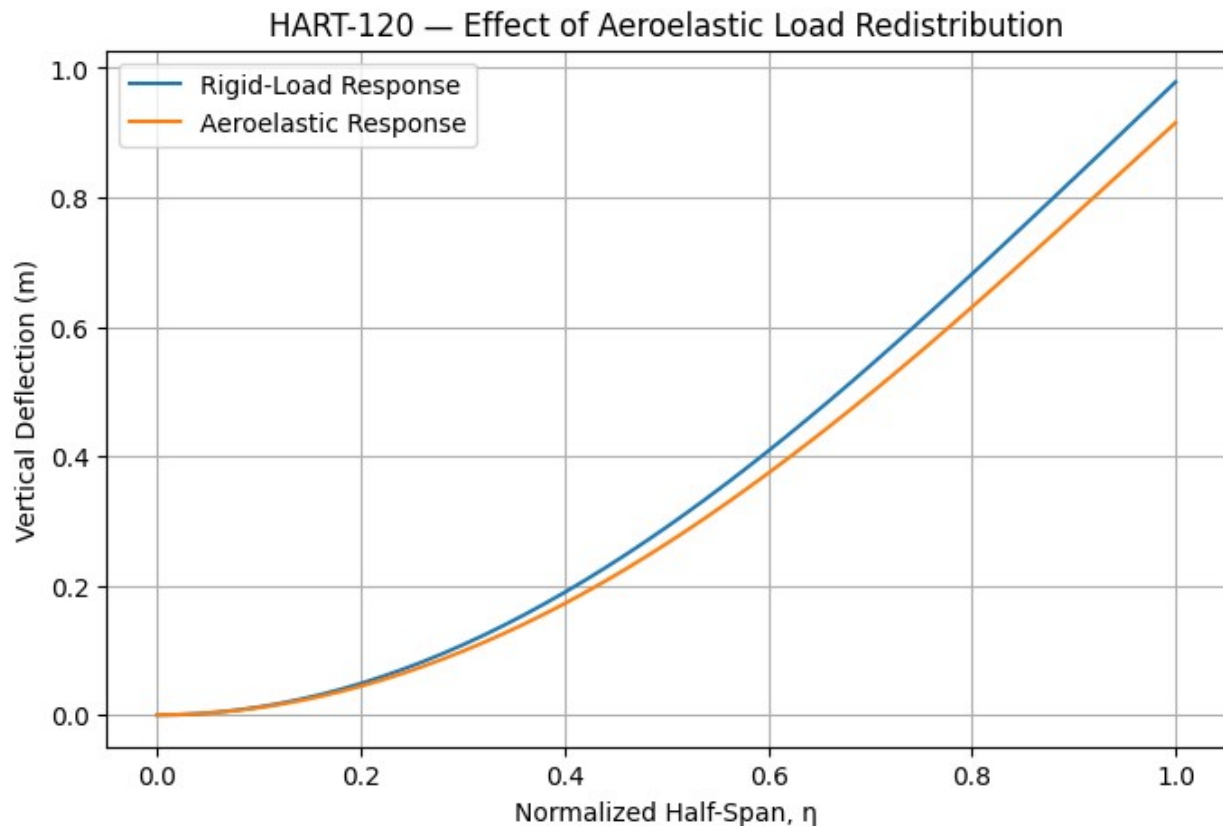


Figure 5. Effect of static aeroelastic load redistribution on selected wing-load metrics.

4.2 Compression-panel buckling screening

A compression-panel buckling study is added after the initial cap sizing. Instead of treating the equivalent spar-cap area as one thick plate, the project reframes the result in terms of allowable panel width-to-thickness ratio and preliminary support spacing.

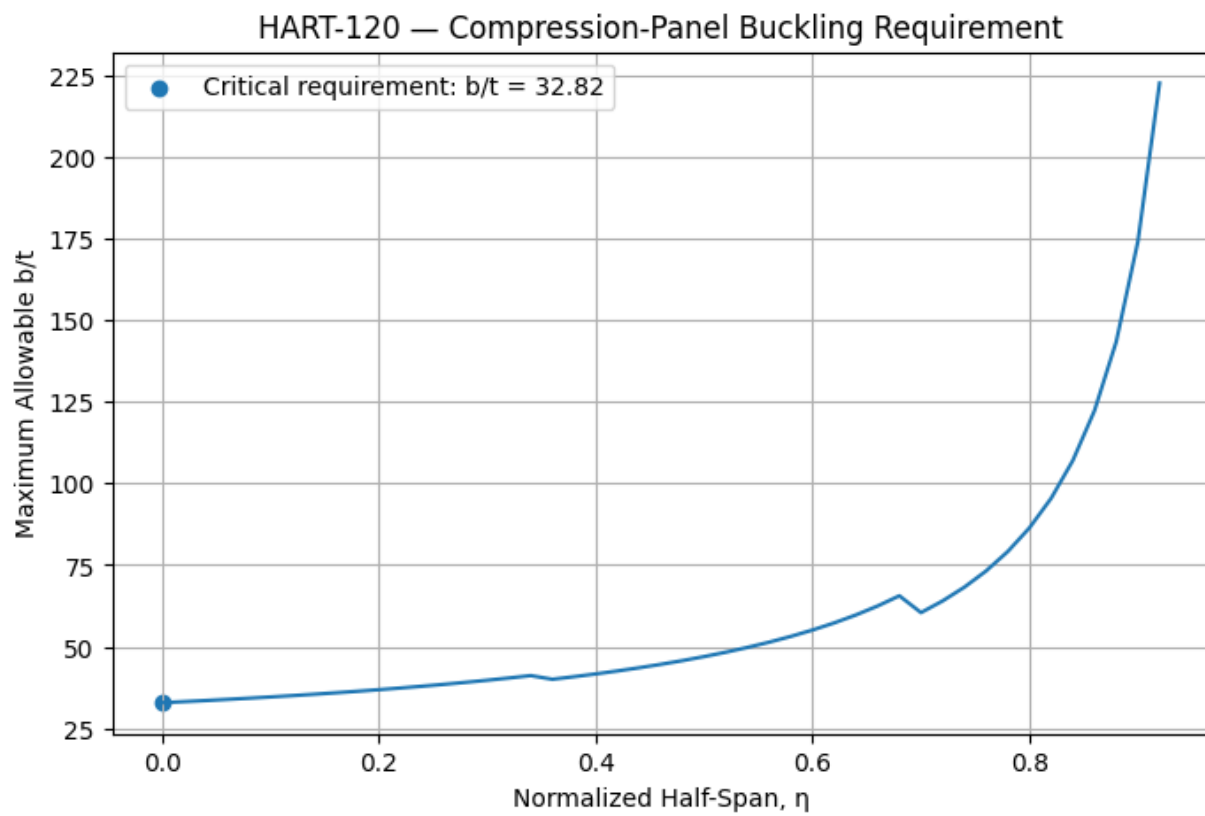


Figure 6. Compression-panel buckling requirement across the structurally relevant span.

5. Aspect-Ratio Aero-Structural Trade

The model then sweeps aspect ratio from 13.5 to 18 at fixed wing area. Increasing AR lowers induced drag, but increases span and root bending demand. Conventional and technology-enabled wingbox designs are compared under common conceptual strength and deflection constraints.

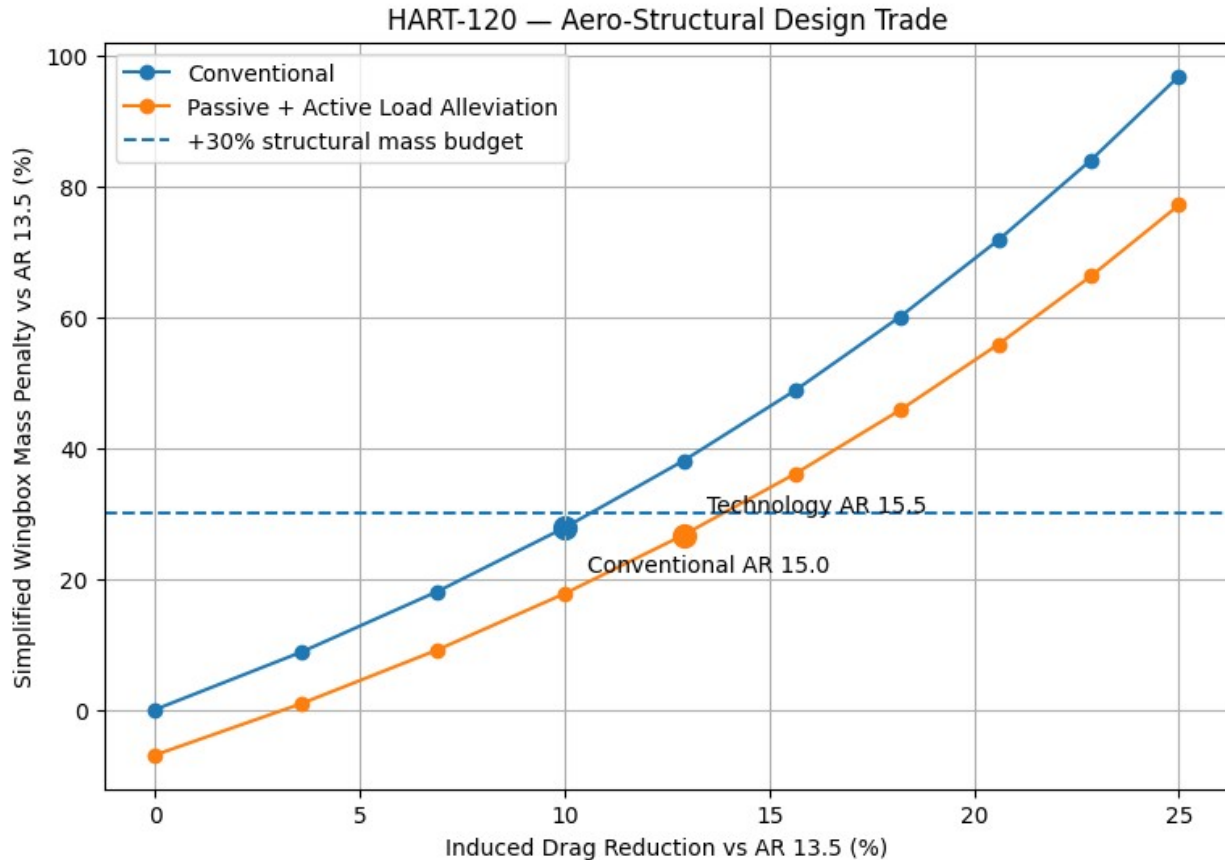


Figure 7. Aero-structural trade: aerodynamic benefit versus simplified wingbox-mass penalty.

Under a +30% structural-mass budget relative to the conventional AR 13.5 reference, the highest conventional configuration is AR 15.0, while passive + active load alleviation permits AR 15.5.

At AR 18 the induced-drag benefit is large, but the simplified structural penalty becomes disproportionate. The result rejects a simplistic 'higher AR is always better' conclusion and motivates a bounded design decision.

6. Degraded Control and Fail-Safe Structural Sizing

The nominal AR 15.5 design is tested with partial and complete loss of maneuver-load-alleviation capability. Spar-cap stress remains below the adopted allowable, but the web shear utilization exceeds unity as control authority degrades.

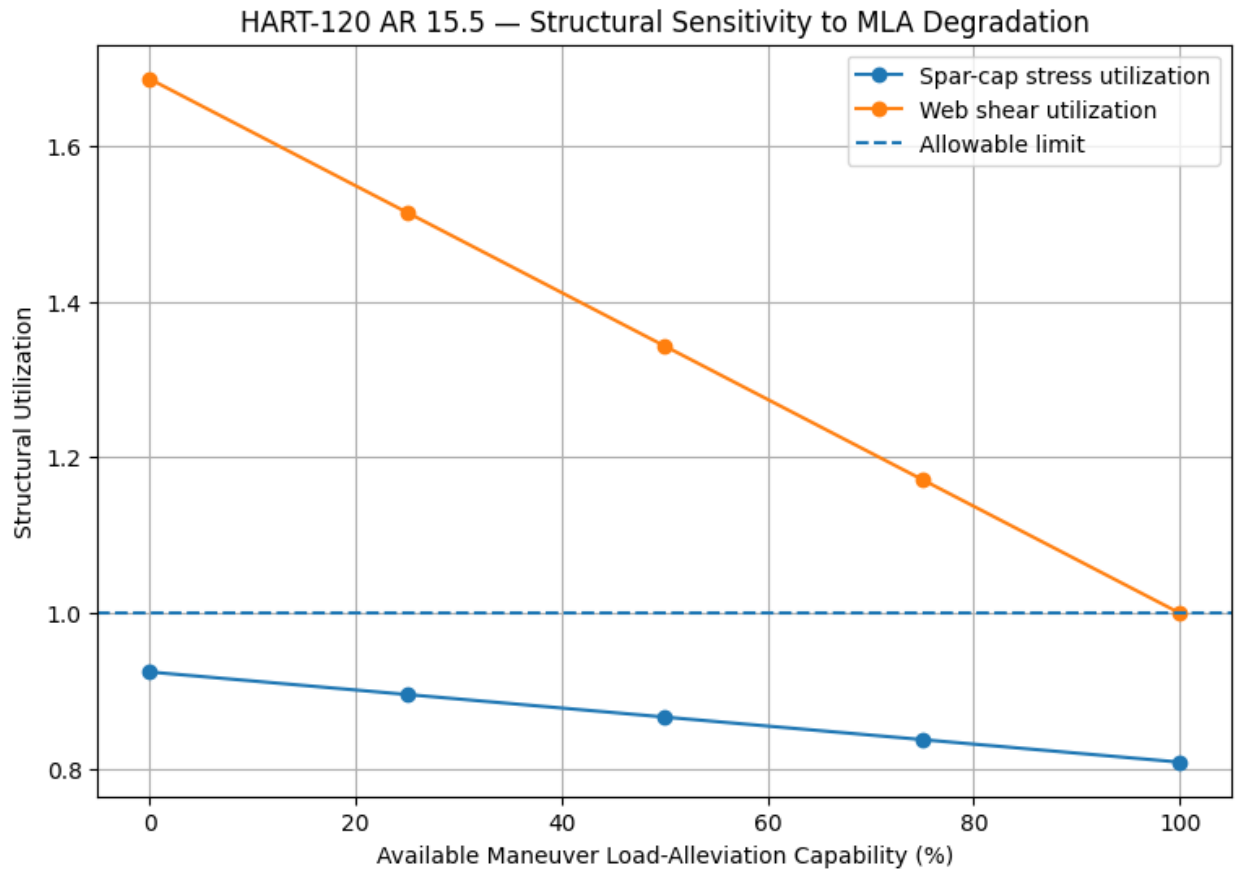


Figure 8. Structural sensitivity of the AR 15.5 concept to degraded maneuver-load alleviation.

The web is therefore resized for the full-control-failure case. This adds 59.66 kg to the fail-safe wingbox, bringing it to 6,143.69 kg.

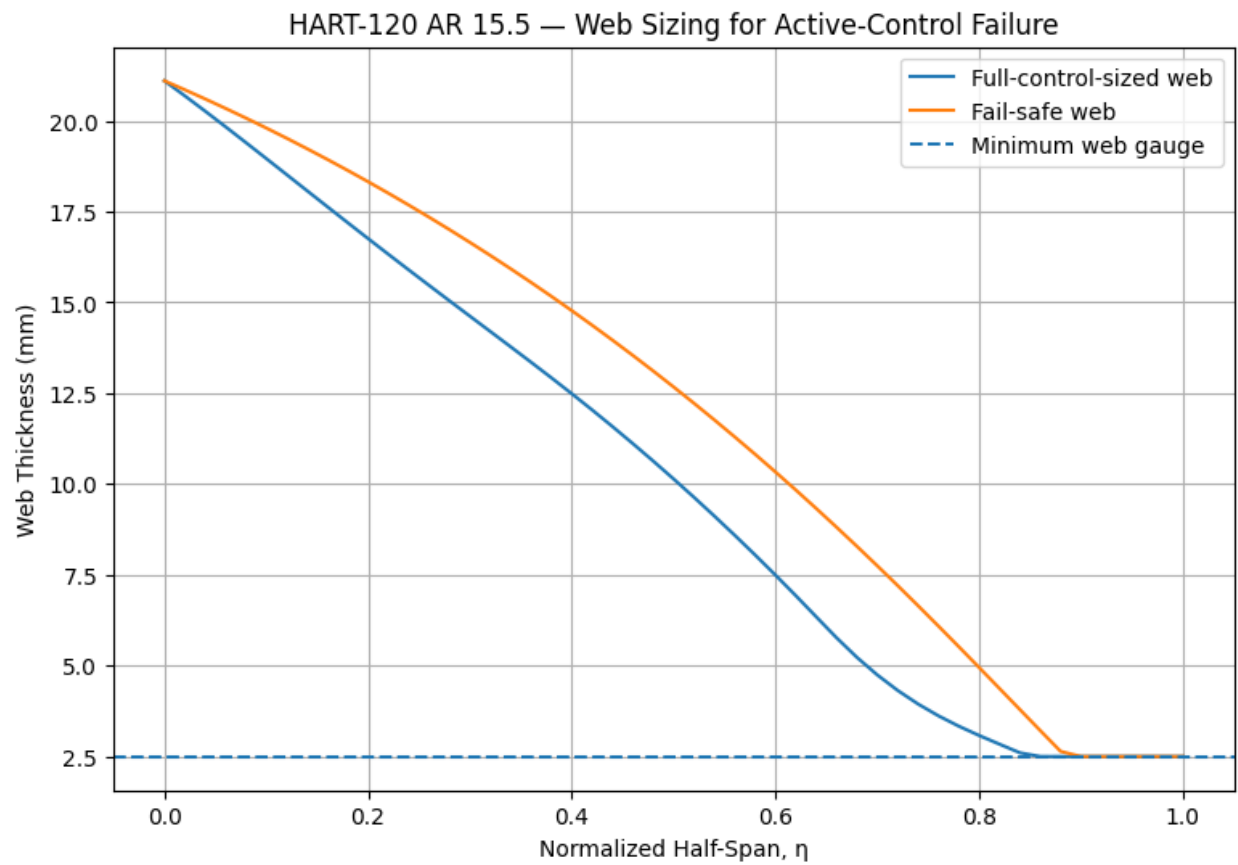


Figure 9. Full-control-sized versus fail-safe web thickness.

6.1 Control-system mass break-even

After fail-safe resizing, 99.16 kg remains before the +30% structural package budget is exceeded. This creates a direct engineering requirement on the total sensing, computing, actuation, wiring and integration mass.

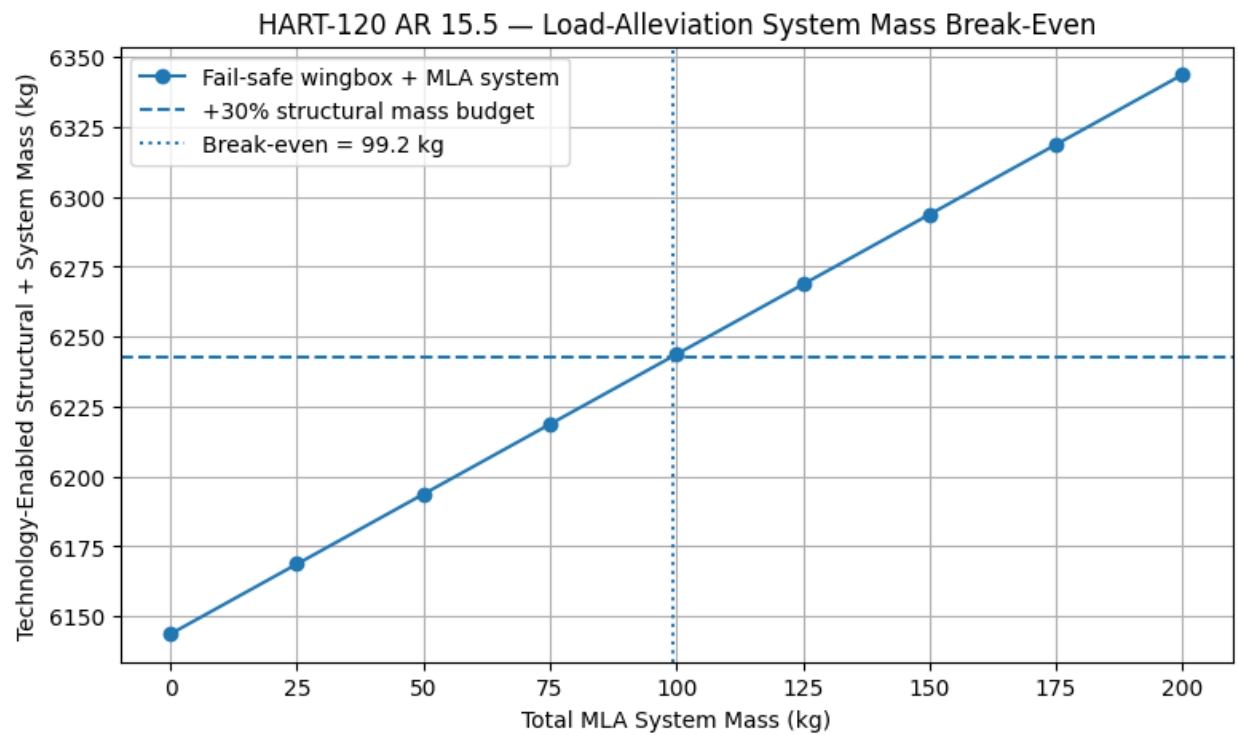


Figure 10. Load-alleviation system mass break-even for the AR 15.5 configuration.

7. Aircraft-Level Aerodynamic and Mission Closure

The final fail-safe wingbox plus a 75 kg conceptual control-system package increases aircraft mass from 55,000 kg to 56,416.5 kg. The heavier aircraft therefore requires a higher cruise lift coefficient, reducing the constant-weight aerodynamic benefit.

After weight coupling, AR 15.5 still reduces induced drag by 8.36% and total cruise drag by 2.28%. L/D improves from 18.91 to 19.85 (+4.97%).

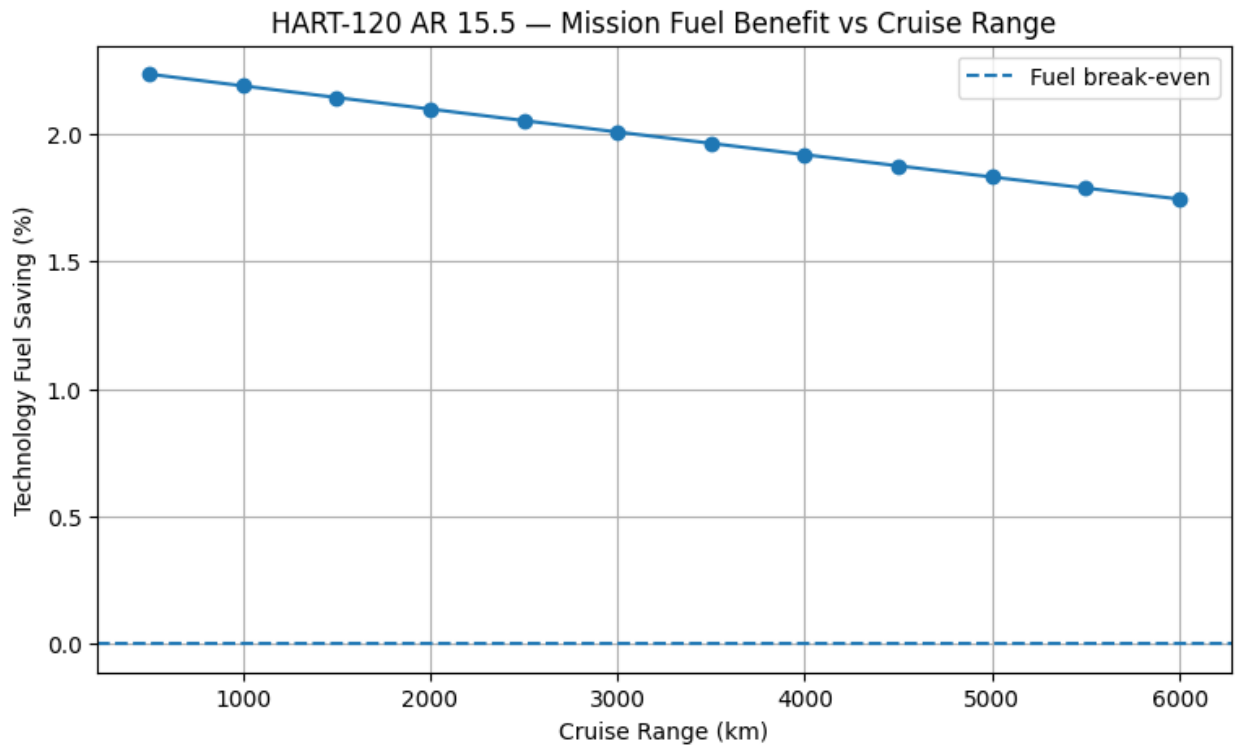


Figure 11. Mission fuel-saving sensitivity with cruise range for the weight-coupled AR 15.5 technology configuration.

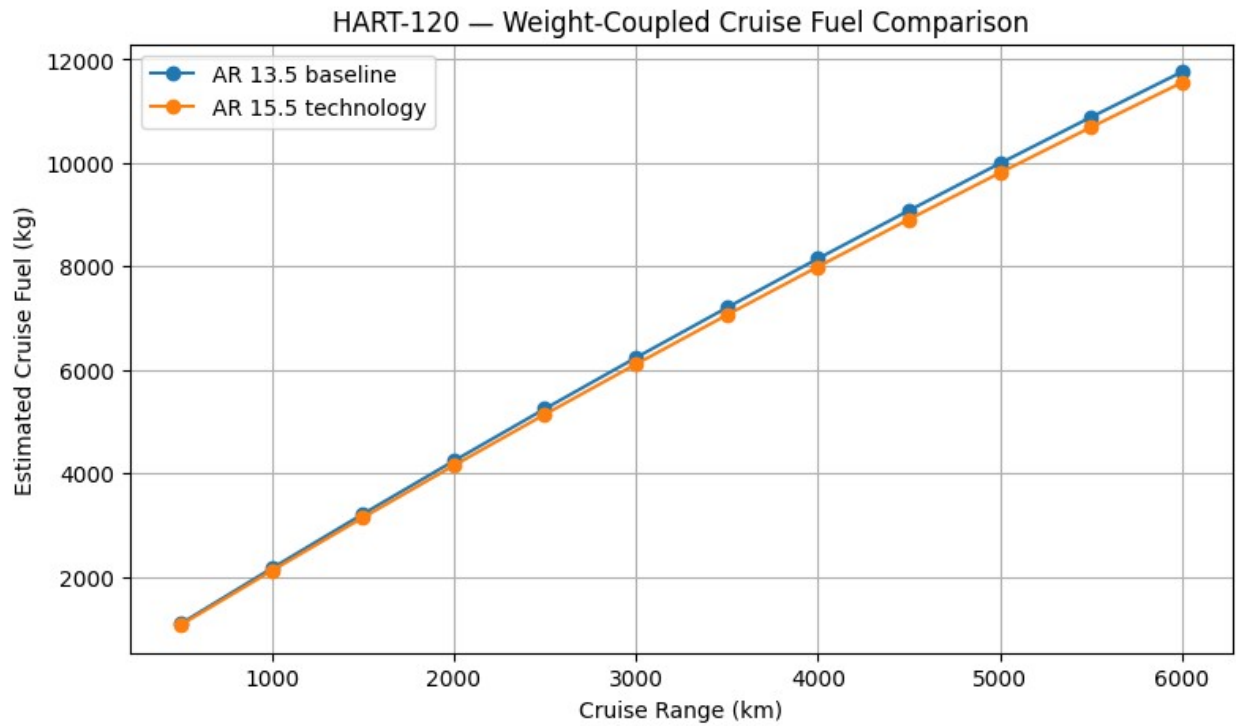


Figure 12. Estimated cruise fuel for the baseline and technology configurations.

For the 3000-km study mission, the reduced-order Breguet-type estimate gives 6,234.17 kg cruise fuel for the baseline and 6,109.06 kg for the technology configuration, a net saving of 125.10 kg or 2.01%.

8. Physics-Generated Machine-Learning Surrogates

Machine learning is introduced only after the physics model is established. The ML models are trained on physics-generated design cases and used to screen large design spaces. They are not treated as replacements for engineering verification.

The multidisciplinary dataset contains 6,000 design cases spanning aspect ratio, control-system mass, parasite drag coefficient and mission range. Random-forest regressors predict fail-safe wingbox mass, cruise drag, L/D, cruise fuel and fuel-saving percentage; a classifier predicts structural feasibility.

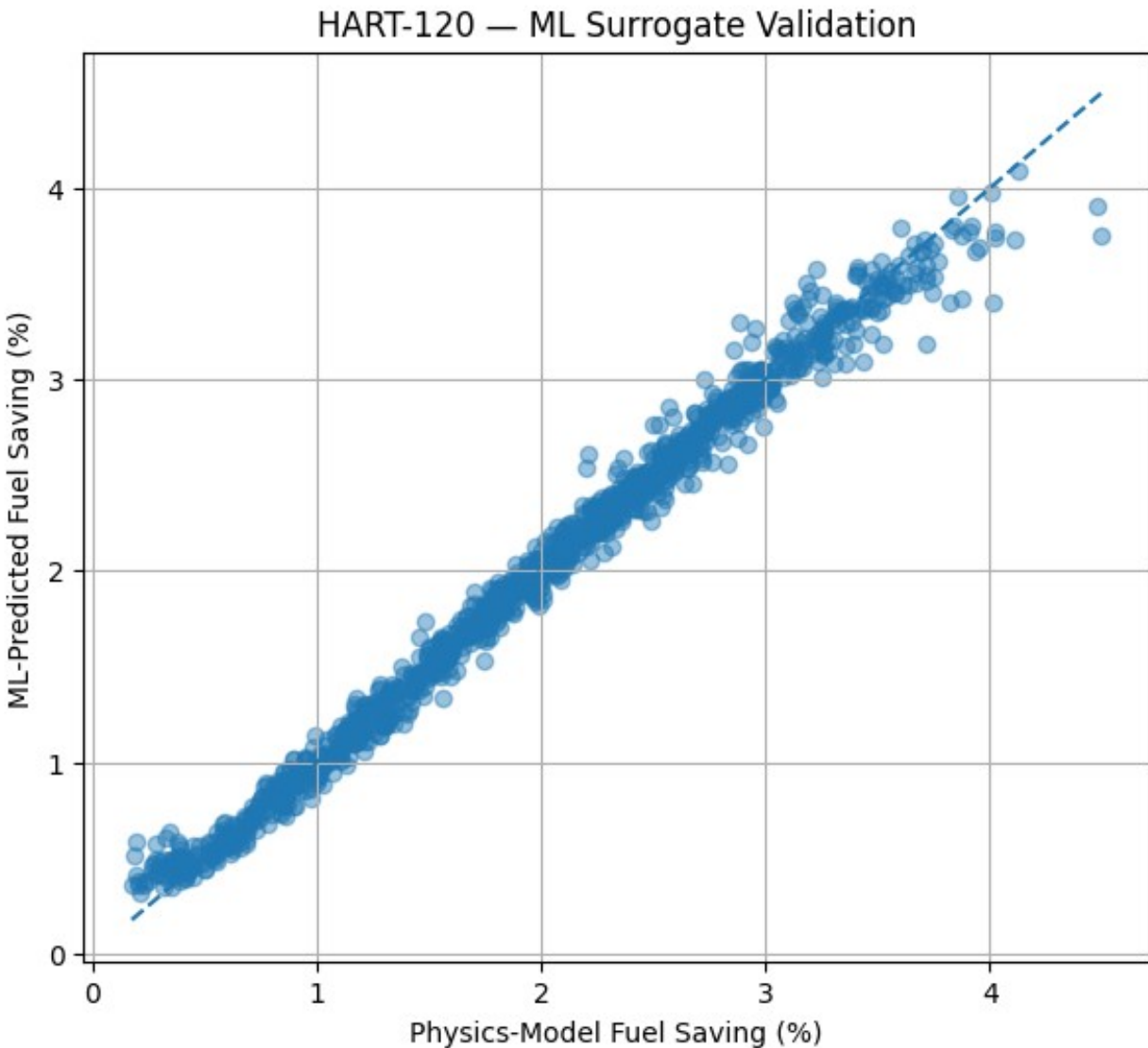


Figure 13. Example ML surrogate validation for cruise fuel-saving percentage.

Selected model performance includes $R^2 \approx 0.997$ for fail-safe wingbox mass, $R^2 \approx 0.998$ for cruise fuel and $R^2 \approx 0.988$ for fuel-saving percentage.

8.1 Dynamic GLA surrogate

A separate 3,000-case physics-generated database is created for dynamic gust-load alleviation. Inputs include gust gradient, actuator time constant, sensor delay, rate capability and preview time. The dynamic

surrogate predicts controlled peak root moment, load reduction, root stress, tip deflection and actuator deflection.

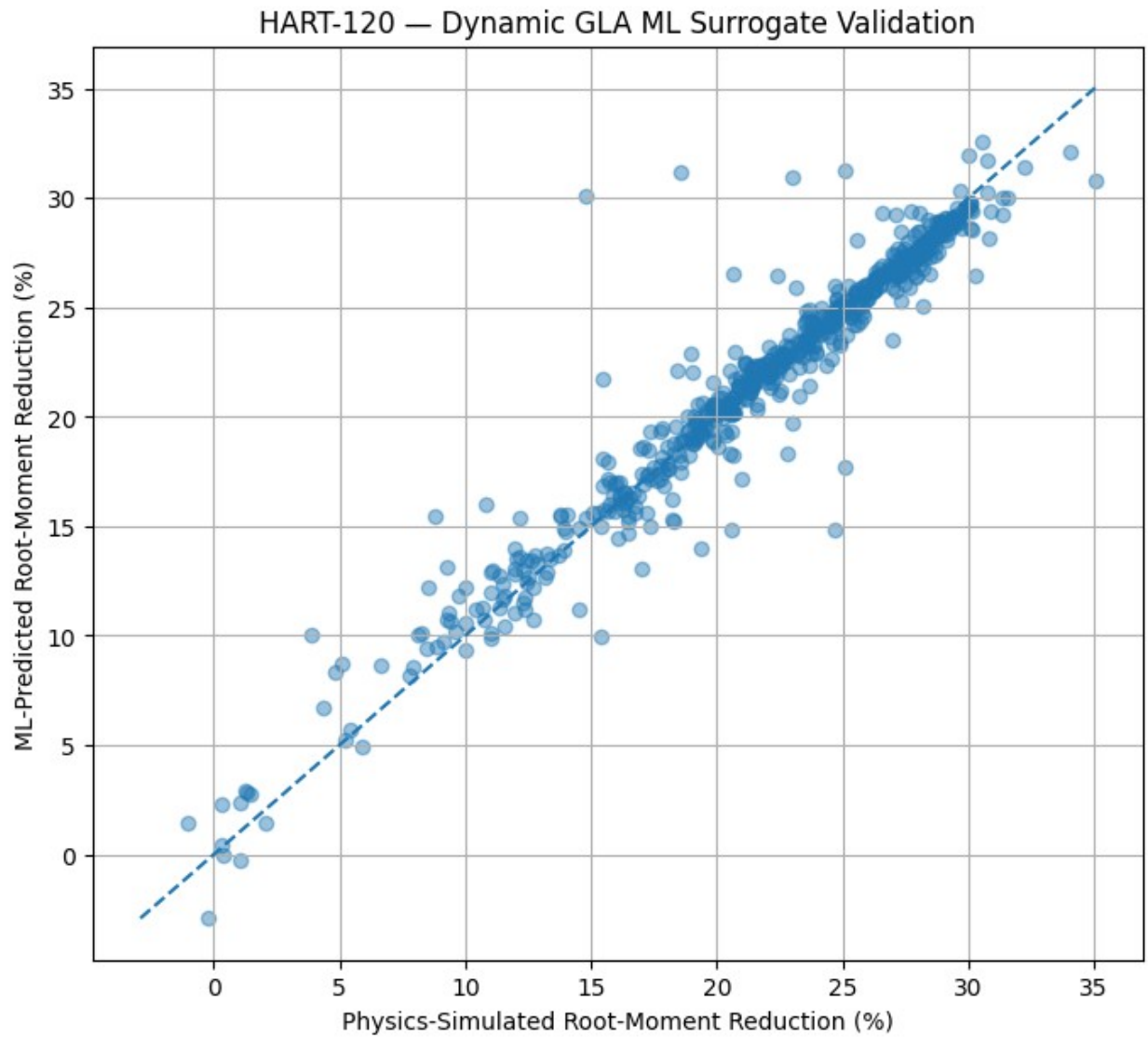


Figure 14. Dynamic GLA surrogate validation against transient physics simulations.

The dynamic root-moment surrogate achieves $R^2 = 0.960$ with an MAE of 0.0415 MN·m; the root-moment-reduction model achieves $R^2 = 0.935$ with MAE ≈ 0.91 percentage points.

9. ML Screening and Full-Physics Re-Verification

The dynamic surrogate screens 100,000 controller/gust design points and then 10,000 controller architectures across 20 gust gradients each. Conservative screening limits are used because the ML surrogate is not exact.

The lowest-burden ML-screened controller is not accepted automatically. The 15 best candidates are returned to the full transient model. Two of the 15 fail the physical controlled-load target, directly demonstrating why surrogate screening must be followed by physics verification.

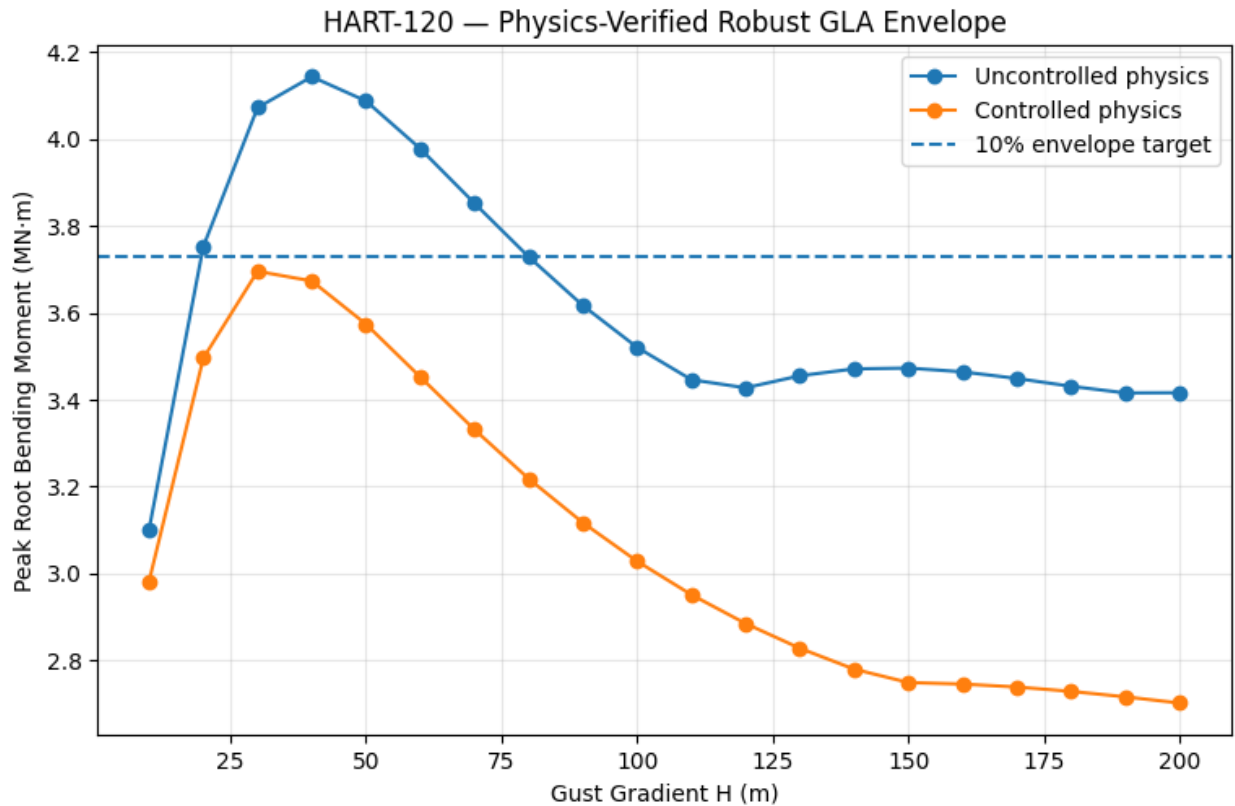


Figure 15. Physics-verified controlled gust envelope for an ML-screened robust controller.

The first physics-verified controller still performs poorly when uncertainty is introduced: its initial 300-case robustness study gives only 57% pass rate against the absolute load target. This becomes a project turning point rather than a hidden failure.

10. Robust Controller Redesign

The uncertainty study shows that preview time, structural frequency and sensor delay strongly influence envelope-level GLA robustness. The robustness criterion is corrected to compare controlled and uncontrolled envelope peaks within the same Monte Carlo realization.

A compact redesign sweep varies preview time and sensor/processing delay. Screening reveals a clear high-robustness region: longer preview and lower delay increase the probability of satisfying both the absolute load limit and the relative $\geq 10\%$ envelope GLA requirement.

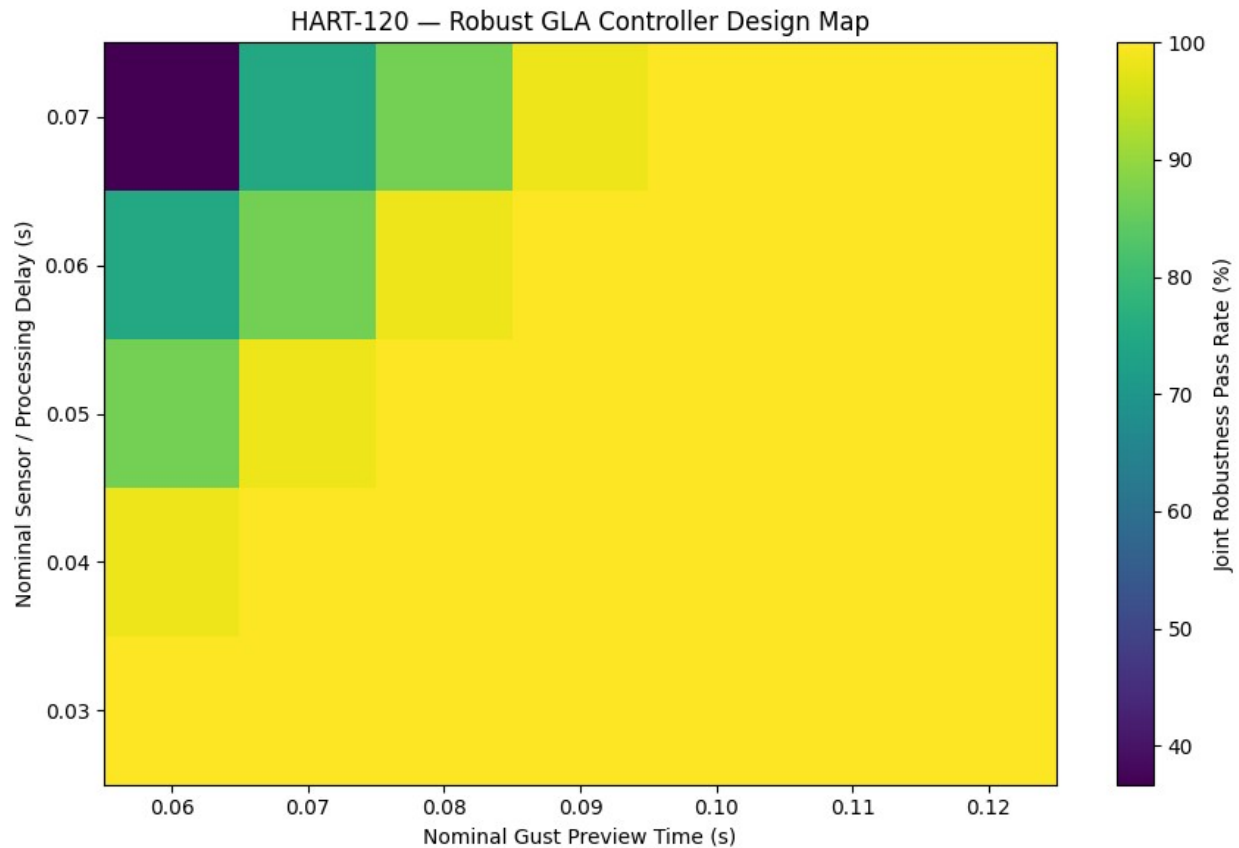


Figure 16. Joint robustness design map used for final controller redesign.

The final controller uses 0.070 s preview and 0.040 s sensor/processing delay, creating 0.030 s effective lead. This is equivalent to approximately 15.69 m of forward sensing at the adopted cruise speed.

11. Technology-System Mass Closure

A conceptual 75 kg GLA/MLA system package is assembled from forward sensing and mounting, flight-control computing, actuator hardware increment, power electronics, wiring/communication and redundancy/integration allowance.

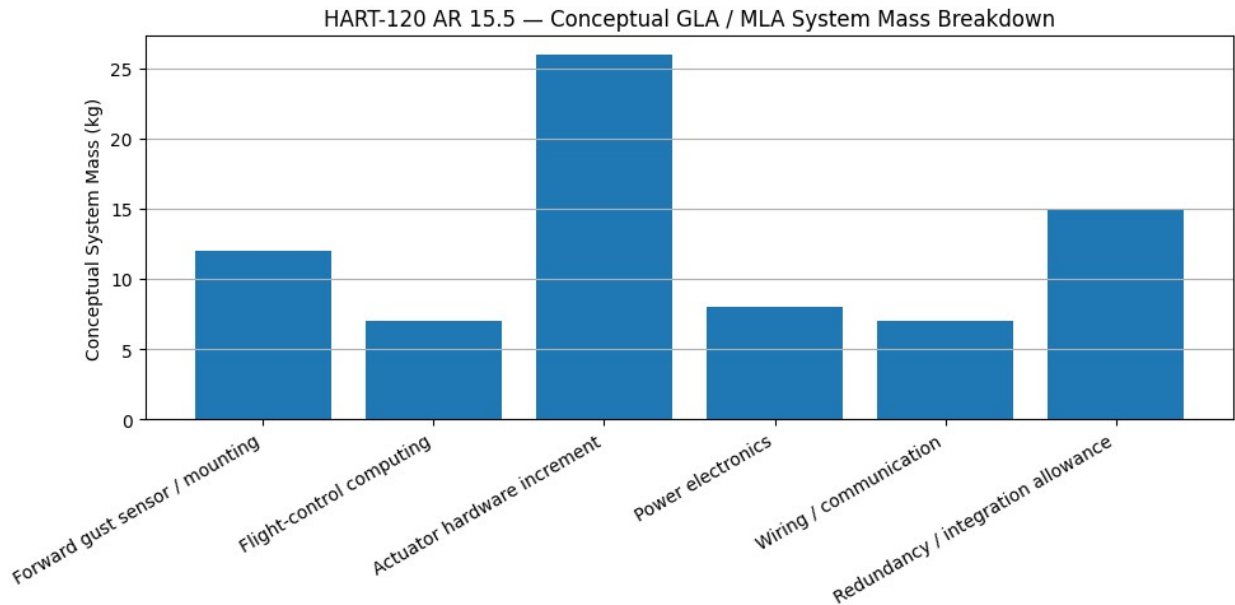


Figure 17. Conceptual mass breakdown of the GLA/MLA system.

With the fail-safe wingbox, the final structural/control package is 6,218.69 kg, leaving 24.16 kg of margin below the 6,242.85 kg project structural budget.

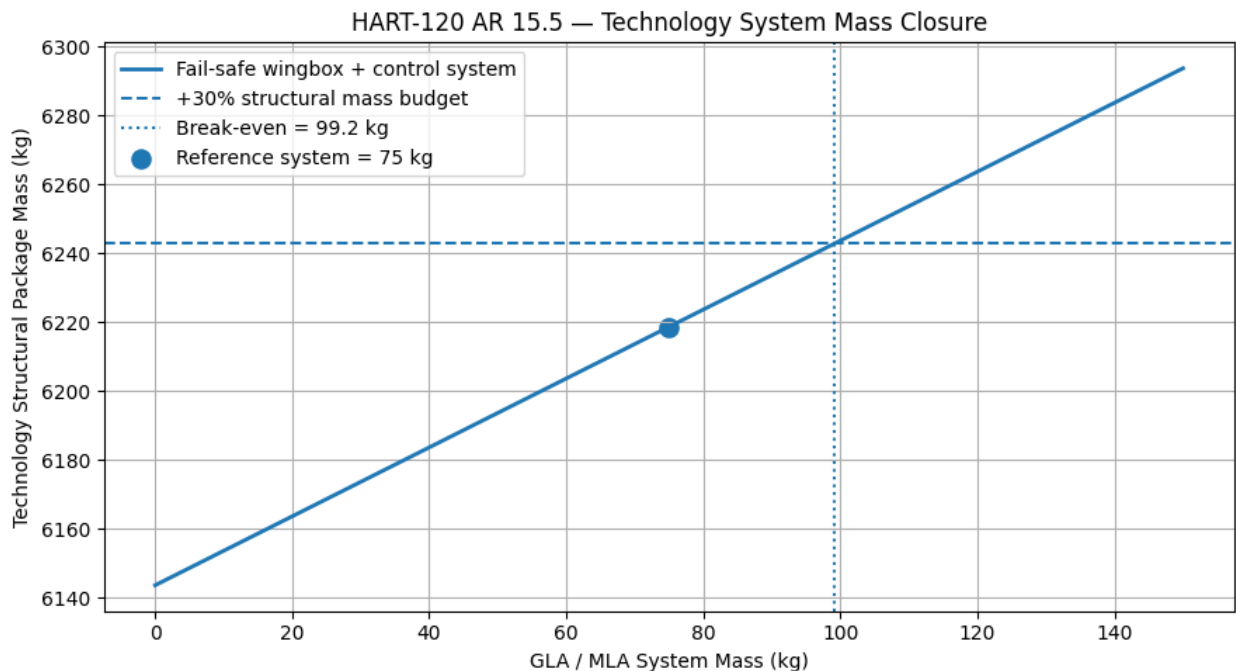


Figure 18. Structural package mass closure and system-mass break-even.

12. Final 300-Case Robustness Verification

The selected 0.070 s / 0.040 s controller is subjected to a final 300-case Monte Carlo analysis. Each realization is evaluated across 20 gust gradients, for 6,000 gust cases in total.

Metric	Baseline / Requirement	Final result
Absolute load pass rate	$\geq 95\%$	98.0%
Envelope $\geq 10\%$ GLA pass rate	$\geq 95\%$	100.0%
Actuator pass rate	$\geq 95\%$	100.0%
Joint robustness	$\geq 95\%$	98.0%
Mean envelope GLA	$\geq 10\%$	15.79%
5th-percentile envelope GLA	$\geq 10\%$	13.81%
95th-percentile controlled load	$\leq 3.729 \text{ MN}\cdot\text{m}$	3.687 MN·m
99th-percentile controlled load	-	3.775 MN·m
Worst observed controlled load	-	3.807 MN·m
Maximum actuator deflection	$\leq 15 \text{ deg}$	13.89 deg

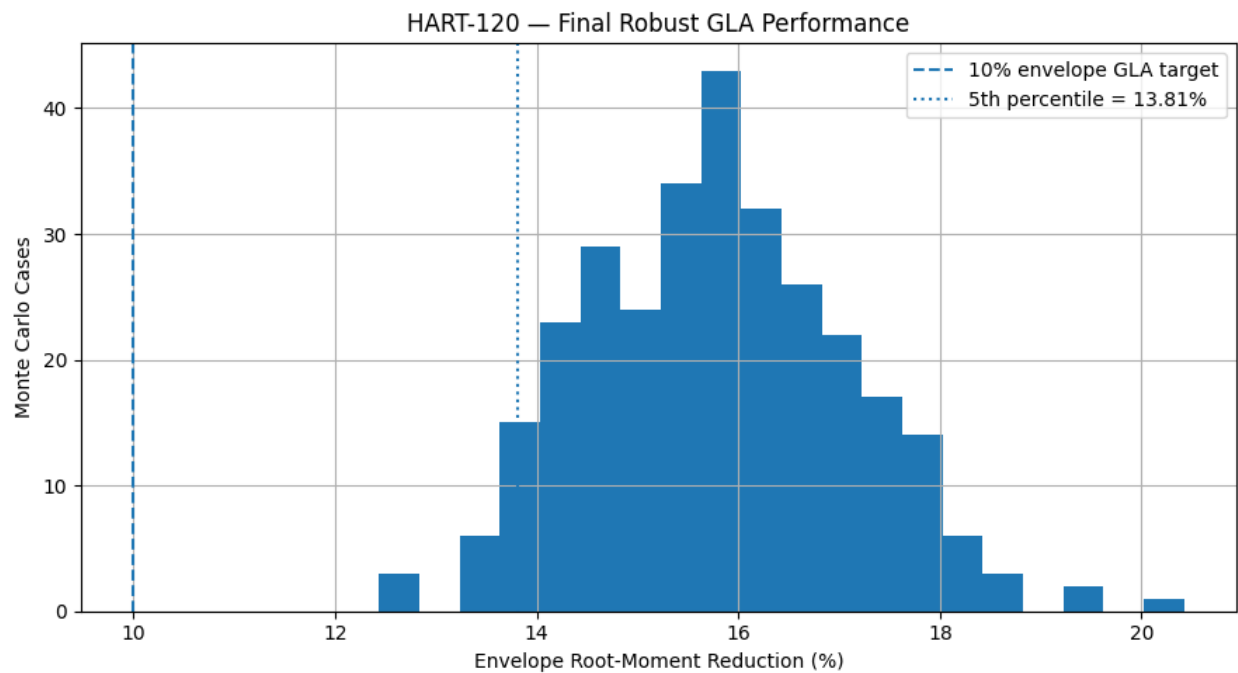


Figure 19. Final envelope root-moment-reduction distribution. The 5th percentile remains above the 10% project target.

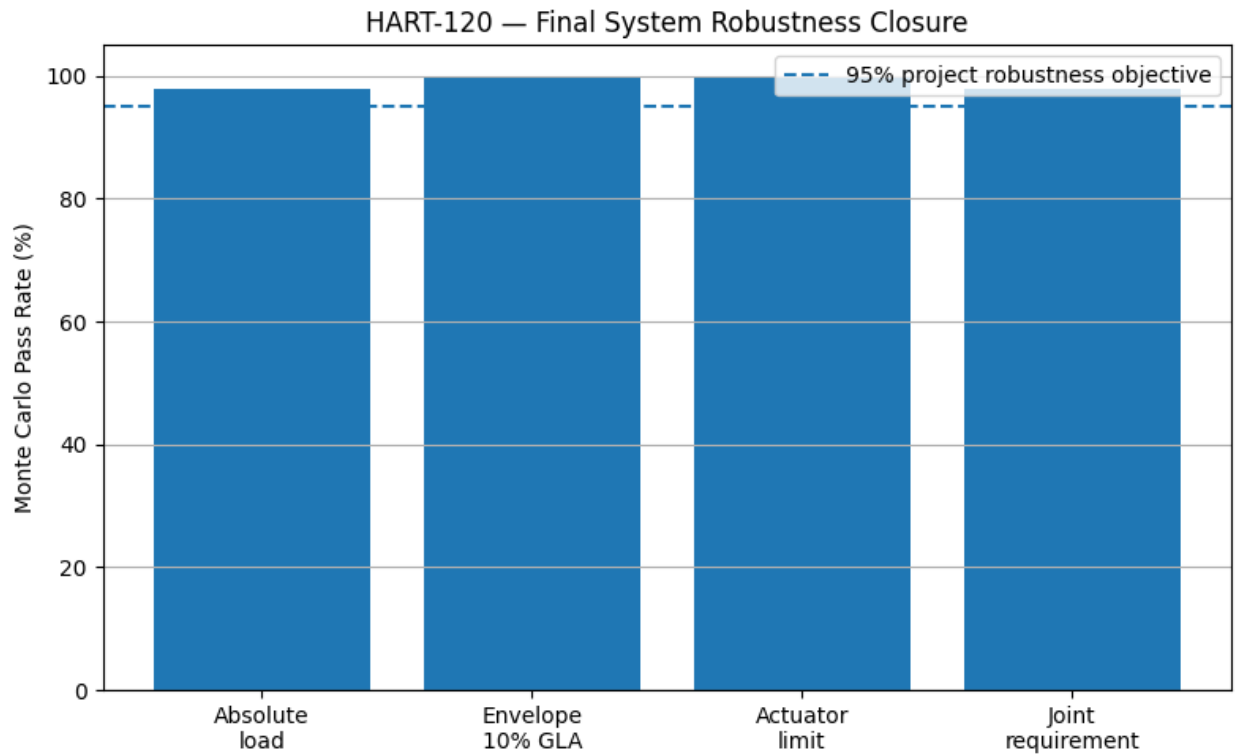


Figure 20. Final project robustness gates.

Final robustness gate: PASS. The controller satisfies the HART-120 project-level robustness objectives within the selected uncertainty ranges.

13. Engineering Interpretation

13.1 What the project solved or meaningfully addressed

- The aerodynamic incentive for higher AR was quantified instead of assumed.
- The corresponding structural mass and bending-load penalty was made explicit.
- Structural material placement was shown to matter as much as total material in the reduced-order stiffness problem.
- Passive aeroelastic washout was linked to spanwise load redistribution.
- Active maneuver/gust alleviation was constrained by control authority, actuator dynamics, delay, rate and deflection limits.
- A nominal load-alleviated structure was challenged with degraded and failed control, exposing a web-shear weakness.
- The fail-safe redesign and control-system mass were included before making an aircraft-level benefit claim.
- The aircraft mass penalty was fed back into cruise lift, drag and fuel estimates.
- ML was used for speed, while final candidates were returned to the physics model.
- A nominally acceptable controller was rejected by uncertainty analysis and then redesigned to reach the project robustness objective.

13.2 What remains unresolved for a real aircraft programme

- 3D transonic aerodynamic effects and shock/boundary-layer interaction.
- Nonlinear aeroelasticity, flutter and divergence clearance.
- Detailed composite laminate sizing, local buckling, joints, fasteners and load introduction.
- Certification-standard discrete-gust and maneuver load envelopes.
- Control-surface hinge moment, actuator power, thermal and redundancy qualification.
- Sensor implementation, latency distribution and failure modes.
- Hardware-in-the-loop and aeroservoelastic control-law validation.
- Complete mission, propulsion, reserve-fuel and operational economics.

13.3 Portfolio value

The project should be presented as evidence of multidisciplinary engineering judgement: defining a design problem, building reduced-order physics, discovering failure modes, refusing to accept an ML prediction without physics re-verification, and closing the final concept against explicit mass, mission and robustness gates.

14. Conclusions

HART-120 began as an aspect-ratio trade and evolved into a system-level conceptual design study because each apparent improvement introduced a new constraint. The final result is therefore not simply 'AR 15.5 is better'; it is a traceable explanation of why that design point remains attractive only after structural, aeroelastic, control, fail-safe, mass, mission and uncertainty effects are included.

- AR 15.5 provides a bounded aerodynamic benefit without the extreme structural penalty seen at higher AR in the adopted model.
- Passive aeroelastic tailoring and active load alleviation recover enough structural design space to move the mass-budget boundary from conventional AR 15.0 to technology-enabled AR 15.5.
- Fail-safe web sizing adds necessary structural mass but does not destroy the AR 15.5 budget closure.
- A 75 kg conceptual GLA/MLA system stays below the 99.16 kg break-even system-mass allowance.
- After aircraft-mass coupling, the final concept retains 8.36% induced-drag reduction, 2.28% total cruise-drag reduction and 2.01% estimated cruise-fuel saving at 3000 km.
- ML accelerates the search but cannot replace the transient model: some ML-screened finalists fail full-physics verification.
- The final preview-based controller reaches 98% joint robustness, with P95 controlled root moment below the project target and 100% pass rate for the relative envelope GLA and actuator constraints.

Overall project status: CLOSED at the conceptual reduced-order level. The study answers the project question with a defensible positive result while making the limitations explicit.

Appendix A. Final Frozen Metrics

Metric	Value
Baseline aspect ratio	13.5
Final aspect ratio	15.5
Baseline span (m)	39.13
Final span (m)	41.93
Baseline aircraft mass (kg)	55000.0
Final aircraft mass (kg)	56416.5
Baseline wingbox mass (kg)	4802.19
Conventional AR15.5 wingbox mass (kg)	6629.17
Nominal technology wingbox mass (kg)	6084.03
Fail-safe wingbox mass (kg)	6143.69
GLA/MLA system mass (kg)	75.0
Final structural-control package (kg)	6218.69
Structural mass budget (kg)	6242.85
Structural mass margin (kg)	24.16
Static aeroelastic tip twist (deg)	-1.769
Induced-drag reduction (%)	8.36
Total cruise-drag reduction (%)	2.28
Baseline L/D	18.91
Technology L/D	19.85
L/D improvement (%)	4.97
Study cruise range (km)	3000
Baseline cruise fuel (kg)	6234.17
Technology cruise fuel (kg)	6109.06
Fuel saving (kg)	125.1

Fuel saving (%)	2.01
Final preview time (s)	0.07
Final sensor delay (s)	0.04
Equivalent sensing distance (m)	15.69
Effective timing lead (s)	0.03
Controlled-load target (MN m)	3.729
Mean envelope GLA (%)	15.79
5th-percentile envelope GLA (%)	13.81
95th-percentile controlled load (MN m)	3.687
99th-percentile controlled load (MN m)	3.775
Worst observed controlled load (MN m)	3.807
Absolute load pass rate (%)	98.0
Envelope $\geq 10\%$ GLA pass rate (%)	100.0
Actuator pass rate (%)	100.0
Joint robustness pass rate (%)	98.0
Max root stress (MPa)	159.87
Max dynamic tip deflection (m)	1.679
Max actuator deflection (deg)	13.89

Source notebook: HART120_Flexible_Wing_Design_FINAL.ipynb

Report status: Project-level conceptual engineering report. Not for certification or production design.